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Torque control strategy for in-wheel motor drive vehicles based on hierarchical deep reinforcement learning considering active safety and energy saving

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Abstract: This study addresses the coordinated optimization of active safety and energy saving for vehicles equipped with in-wheel motors, overcoming the limitations of existing torque distribution methods that exhibit incomplete energy-saving objectives and weak multi-objective coordination. A hierarchical deep reinforcement learning approach based on a soft actor-critic algorithm is proposed, in which electric drive and tire slip losses are uniformly quantified and integrated into a unified objective. The framework uses a two-level architecture: an upper-level agent determines the total driving torque and reference yaw moment for longitudinal and yaw stability regulation, while a lower-level agent optimizes individual wheel torque allocation by adaptively balancing safety and energy saving. Validation shows that hierarchical decomposition substantially improves training stability and convergence versus single-agent methods. Under double lane change maneuvers, the method achieved a longitudinal speed deviation below 0.60 km/h and a maximum lateral displacement error of 0.20 m while effectively constraining the yaw rate and sideslip angle. Step steering tests confirmed robust generalization. Extended validation on a 5-km 3D road demonstrated enhanced active safety while simultaneously achieving energy loss reductions from 6.62% to 16.77%. Hardware-in-the-loop tests under ±10% mass perturbations revealed consistent performance across all dynamic metrics between real-time and simulation environments, with energy saving deviations confined to within 2%. The approach provides a practical foundation for intelligent coordinated control in future electric vehicles.

Key words: In-wheel motor drive vehicles; Hierarchical deep reinforcement learning; Active safety; Energy saving

1 Introduction

Amid mounting environmental concerns and energy crises, electric vehicles have experienced unprecedented development opportunities. Unlike conventional centralized drive systems, in-wheel motor drive vehicles (IWMs, Fig. 1) feature a more flexible powertrain layout that enhances overall vehicle dynamic performance and control accuracy (He and Hori, 2006; Hang et al., 2017; Tian et al., 2019; Hang and Chen, 2021), demonstrating significant technical advantages for active safety control, energy-saving control, and rapid response capabilities.

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In recent years, active stability control for IWMs has been a prominent research focus. Most studies have been based on an ideal two-degree-of-freedom (2-DOF) model and use fuzzy control (Tahami et al., 2004; Zhai et al., 2016; Silva et al., 2022), model predictive control (Shi et al., 2018; Zha et al., 2021; Zhai et al., 2022), or sliding mode control (Canale et al., 2009; Zhai et al., 2016; Ding et al., 2017; Hu et al., 2020; de Carvalho Pinheiro et al., 2021; Gao et al., 2024) methods to compute the desired yaw moment.

The lower-level controller formulates torque allocation by mapping the total propulsion demand and reference yaw moment to individual in-wheel motors, subject to stability constraints. These studies have significantly advanced stability control for IWMs, but most rely on model-based control approaches that entail inherent complexity. The nonlinear characteristics of tire models render control strategy design more challenging and online optimization computationally

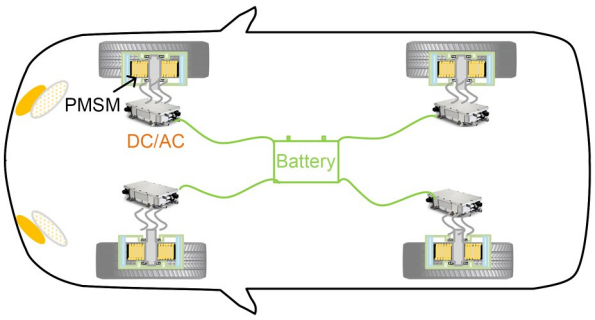


Fig. 1 Architecture of IWMs. PMSM: permanent magnet synchronous motor; DC: direct current; AC: alternating current

intractable (Ma et al., 2019). Furthermore, parametric uncertainties in vehicle dynamics can degrade control performance and even compromise the robustness and generalization capability of control strategies (Peng et al., 2020).

Energy saving has also emerged as a prominent research focus in torque control for IWMs. Many scholars have approached this challenge from the perspective of enhancing the efficiency of electric drive systems. Pennycott et al. (2015) investigated the energy-saving potential of reducing motor unit power losses through wheel torque allocation. Shen et al. (2016) proposed a proportional allocation method based on computed power loss and efficiency maps to improve vehicle energy saving. Hu et al. (2019) analyzed the influence of different wheel torque distribution schemes on vehicle stability and energy-saving performance during steering maneuvers, and subsequently developed an energy-saving optimal torque allocation strategy for steering scenarios. Li et al. (2020) combined motor efficiency maps with a quadratic programming (QP) algorithm to enhance energy-saving performance. Energy losses induced by tire slip constitute a critical aspect of the optimization process, particularly because torque distribution significantly influences power dissipation caused by tire slip (Pennycott et al., 2015). Researchers have proposed numerous methods, including model predictive control (Yuan et al., 2015; Zhang et al., 2024), feedforward and feedback control (Jia et al., 2018), and sliding mode control (de Carvalho Pinheiro et al., 2021; Kang et al., 2023; Gao et al., 2024), to either regulate the tire slip ratio within an optimal range or design control strategies for distributed drive vehicles with tire slip energy as the control objective. However, under certain conditions, conflicts between safety and energy-saving control objectives arise in IWMs, fundamentally reflecting a dynamic trade-off between friction circle utilization and energy loss. Current research faces several limitations and challenges: studies mostly adopt single-objective optimization or rule-based switching between objectives, leading to transient conflicts and constrained optimization performance; energy saving efforts frequently focus on electric drive efficiency while neglecting tire slip energy loss—a critical component strongly coupled with torque vectoring. Confronted with the strong nonlinearity and parametric uncertainties of tire–road interactions, model-based approaches encounter difficulties in meeting real-time constraints and managing decoupling complexity. Moreover, the prevalent reliance on model-based control methods introduces increased control complexity stemming from model nonlinearities and uncertainties, rendering online optimization computationally intractable.

Deep reinforcement learning (DRL) integrates sequential decision-making with function approximation to address complex control tasks. As a paradigm, it shows considerable potential for controlling nonlinear systems and has been increasingly applied to vehicle control in recent years (Tang et al., 2021; Wei et al., 2022a, 2022b; Cui et al., 2023; Ma et al., 2026). DRL methods explore optimal control trajectories through iterative optimization, effectively addressing control challenges arising from model complexity and nonlinearity (Koh et al., 2020; Li et al., 2020; Khani et al., 2024). Among various DRL algorithmic implementations, the soft actor-critic (SAC) algorithm is recognized for its proficiency in navigating high-dimensional action spaces under competing objective constraints, leading to its broad adoption in complex control applications (Sun et al., 2022; Vohra et al., 2022; Chen et al., 2026). However, when applying DRL to controlled systems with complex control objectives and high-dimensional state-action spaces, the design of reward functions poses significant challenges, often leading to poor convergence during training. In the field of vehicle dynamics control, hierarchical methods have been widely adopted to decompose high-order nonlinear problems into several low-dimensional subproblems, thereby reducing computational complexity and enhancing real-time performance under multiple constraints. Among the various hierarchical control frameworks, approaches based on reinforcement learning (RL) have

recently gained increasing attention. To provide a focused review, this study concentrated on the application of RL, specifically hierarchical deep reinforcement learning (HDRL) methods, in vehicle dynamics control, as this is the domain most relevant to our work. Within this scope, most studies remain confined to hybrid architectures that combine RL with traditional control algorithms, rather than using fully data-driven strategies. For instance, Ma et al. (2024) proposed a hierarchical longitudinal control system that integrates the deep deterministic policy gradient (DDPG) algorithm with proportional-integral-derivative (PID) control, utilizing RL to dynamically tune PID parameters in real time to accommodate variations in vehicle dynamics. Similarly, Peng et al. (2025) designed a hierarchical trajectory tracking framework tuned by DRL, integrating the twin delayed deep deterministic policy gradient (TD3) algorithm with an external attention mechanism to enhance the performance of kinematic model predictive control (MPC) in uncertain environments. Although these studies have advanced hierarchical vehicle control to various degrees, they remain inherently dependent on traditional control laws or specific dynamic models. Such reliance restricts the exploration space and generalization potential of agents in end-to-end decision-making processes. Beyond vehicle dynamics control, HDRL methods have also been introduced into the domain of autonomous driving decision-making. Addressing the complex temporal task of overtaking, Lodhi et al. (2026) used a master policy to schedule sub-task agents for both lane changing and straight-line driving, effectively mitigating the low training efficiency of single-agent systems in long-horizon complex maneuvers. Similarly, targeting autonomous navigation tasks, Yang et al. (2026) decoupled multi-variable vehicle acceleration and steering control into hierarchical sub-tasks using DRL, which significantly improved policy optimization efficiency. However, these studies operate at the decision-making and path-planning level, which is distinct from the low-level vehicle dynamics coordination problem addressed in this work. A critical research gap thus remains: despite the demonstrated potential of HDRL architectures in high-level decision-making tasks, few studies have extended this paradigm to the coordinated control of vehicle dynamics—particularly for IWMs—to simultaneously optimize active safety and energy efficiency. Existing DRL-based approaches often overlook the complex coupling relationships among electric drive losses, tire slip dynamics, and active safety requirements. Moreover, when DRL methods are directly applied to the high-dimensional state-action spaces of IWMs, they frequently suffer from the "curse of dimensionality", resulting in suboptimal training stability. To the best of our knowledge, no study has addressed this specific challenge through an HDRL framework that integrates unified quantification of electric drive and tire slip losses with active safety coordination. This gap motivated the present study.

To address the above challenges, we propose a control method termed HDRL control for the coordinated optimization of active safety and energy saving in IWMs. The main innovations of this study are threefold: first, active safety and energy saving are jointly considered by uniformly quantifying electric drive losses and tire slip losses and incorporating them into a unified optimization objective; second, DRL is introduced to achieve simultaneous optimization of active safety and energy saving for IWMs; third, to mitigate training difficulties arising from complex control objectives and high-dimensional state-action spaces, the control objectives are hierarchically decomposed based on the SAC algorithm, establishing an HDRL framework that enhances training stability and convergence speed.

2 Algorithmic framework for DRL

The proposed control architecture (Fig. 2) adopts an HDRL framework. The upper agent outputs the total driving torque and reference yaw moment to regulate the longitudinal speed and yaw stability. The lower agent, through a carefully designed multi-objective reward function, balances active safety and energy saving, where energy saving accounts for both electric drive losses and tire slip losses. Both agents are implemented using the SAC algorithm and interact with the vehicle dynamics model as the environment. This hierarchical mechanism enables coordinated optimization of global vehicle dynamics and safety–energy trade-offs. The following section details the vehicle model and control design.

2.1 Vehicle system modeling

2.1.1 Vehicle system dynamics

This study used the classical 2-DOF vehicle model to describe the lateral–yaw dynamics (Fig. 3), neglecting

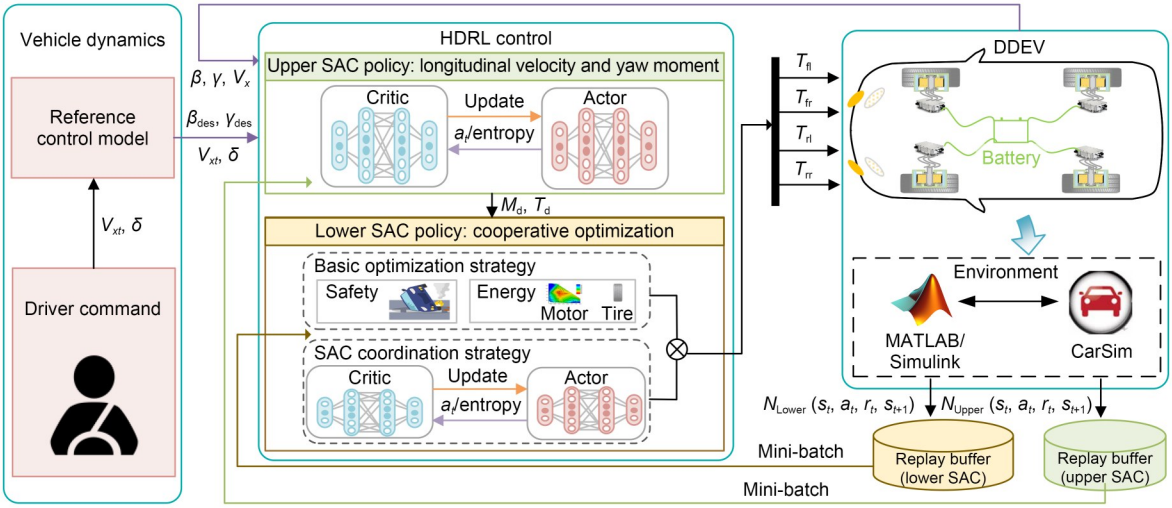


Fig. 2 Control architecture proposed in this paper. N_{Lower} is the experience transition tuple generated by the lower SAC agent; N_{Upper} is the experience transition tuple generated by the upper SAC agent; the other variables in this figure are explained in the text

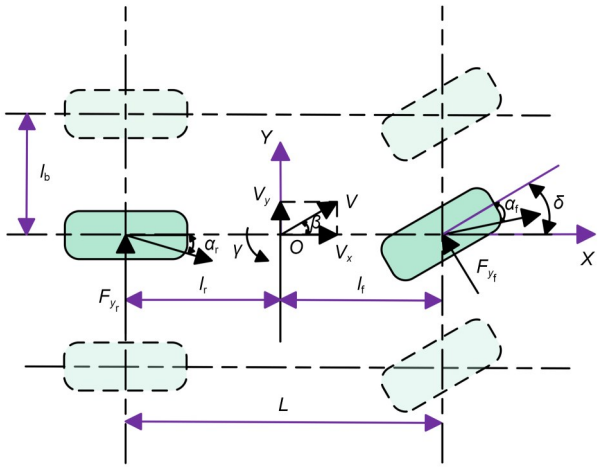


Fig. 3 Illustration of vehicle system dynamics. l_b is half of the wheel track; V is the vehicle velocity; V_y is the vehicle lateral velocity; L is the wheelbase

pitch and roll motions. This simplification was justified because the control objectives—longitudinal speed tracking, yaw stability regulation, and energy-efficient torque allocation—do not depend on pitch or roll feedback. Moreover, the safety metrics (sideslip angle and yaw rate) are defined in the horizontal plane and adequately represent yaw stability. The model is valid for low-to-moderate speeds and for small sideslip angles. Although its fidelity diminishes under extreme steering or tire saturation, this limitation is partially compensated by the subsequent incorporation of the Magic Formula tire model, extending the applicability to low-adhesion surfaces. The governing equations for the sideslip angle and yaw rate dynamics are given by Abe (2015) and Wei et al. (2022b):

$$\begin{cases} mV_x(\dot{\beta}+\gamma)=F_{y_f}\cos\delta+F_{y_r}, \\ I_z\dot{\gamma}=l_fF_{y_f}\cos\delta-l_rF_{y_r}, \end{cases} \tag{1}$$

where m denotes the total vehicle mass; V_x is the longitudinal velocity; β represents the sideslip angle; γ is the yaw rate; I_z denotes the moment of inertia about the z-axis; δ is the front wheel steering angle; F_{y_f} and F_{y_r} represent the lateral forces of the front and rear wheels, respectively; l_f and l_r represent the longitudinal offsets of the vehicle's center of gravity relative to the front and rear axles, respectively. The lateral tire forces are nonlinear functions of the slip angles, which are expressed as:

$$\begin{cases} \alpha_f=\beta+\dfrac{l_f\gamma}{V_x}-\delta, \\ \alpha_r=\beta-\dfrac{l_r\gamma}{V_x}, \end{cases} \tag{2}$$

where α_f and α_r denote the slip angles of the front and rear tires, respectively.

2.1.2 Reference dynamic model

By imposing steady-state conditions on the vehicle dynamics described in Eqs. (1) and (2), i.e., neglecting the temporal derivatives of the state variables, the

equilibrium references for the sideslip angle and yaw rate were obtained. The detailed derivations and formulations of the reference model are provided in Section S1.1 of the electronic supplementary materials (ESM).

2.1.3 Magic Formula tire model

Tires are the sole vehicle–road contact components, and their slip-related energy losses are critical for energy saving quantification. In this study, the Magic Formula was adopted to characterize tire force generation due to its unified structure and high fitting accuracy. The generic expression of the Magic Formula, along with the definition of the longitudinal slip ratio, is provided in Section S1.2 of the ESM. The parameters used in the model follow the standard formulation described therein.

2.1.4 Power loss analysis of motor and modeling

Motor efficiency serves as the basis for calculating motor energy losses. The vehicle was equipped with four identical in-wheel motors; the corresponding efficiency map and key motor parameters are provided in Section S1.3 of the ESM. The power loss of a single motor is calculated as:

$$P_{\text{loss},i}=\frac{T_{\text{m},i}n_i(1-\eta_i)}{9549\eta_i}, \tag{3}$$

where n_i denotes the rotational speed of the motor at wheel i; $T_{\text{m},i}$ represents the torque of the motor at wheel i; η_i is the efficiency of the motor at wheel i.

2.1.5 Power loss analysis of tire slip and modeling

Tire slip power loss (P_{tireloss}) originates mainly from heat dissipation induced by tire sliding relative to the road surface, encompassing both longitudinal slip energy loss and lateral slip energy loss (Guo et al., 2020; Xu et al., 2023):

$$\begin{cases} P_{\text{tireloss}}=P_{\text{lateral}}+P_{\text{longitudinal}}, \\ P_{\text{lateral}}=\sum_{i=1}^{4}(-F_{y_i}v_{x_i}\sin\beta_i), \\ P_{\text{longitudinal}}=\sum_{i=1}^{4}F_{x_i}\lambda_{x_i}v_{x_i}\cos\beta_i, \\ \sum E_{\text{tireloss}}=\int(P_{\text{lateral}}+P_{\text{longitudinal}})\,\mathrm{d}t, \end{cases} \tag{4}$$

where P_{lateral} denotes the lateral slip power loss; $P_{\text{longitudinal}}$ represents the longitudinal slip power loss; $\sum E_{\text{tireloss}}$ is the total tire slip energy loss; β_i denotes the lateral angle of wheel i; v_{x_i} denotes the longitudinal velocity at the center of wheel i in the tire coordinate system.

2.2 Description of the reinforcement learning algorithm system

DRL abstracts the closed-loop control problem as a Markov decision process (MDP), whose detailed mathematical formulations are provided in Section S2.1 of the ESM.

When applying DRL to vehicle control, the dimensionality of the action space critically affects algorithmic convergence and performance. Directly mapping states to four-wheel torques poses a high-dimensional policy search challenge, where the required sample size grows exponentially with action dimensionality, often leading to unstable training. To address this, our HDRL framework uses separation of concerns, decoupling the complex four-dimensional torque allocation problem into two low-dimensional subtasks: the upper-level agent outputs a 2D control vector (total driving torque and reference yaw moment) to satisfy vehicle kinematics, while the lower-level agent dynamically balances safety and energy saving via a scalar weight coefficient. The final torque distribution is then solved by a QP solver. This hierarchical architecture reduces the policy search space, accelerates training convergence, and inherently incorporates physical constraints, thereby enhancing the practicality and reliability of DRL in complex control systems.

2.2.1 Upper agent for integrated longitudinal yaw control

For the upper agent, the objective is to maintain vehicle longitudinal speed and yaw stability by outputting the total driving torque and target yaw moment required by the vehicle. Since the state space provides the input information for RL and forms the basis for agent decision-making, sufficient critical information must be provided to the agent. Accordingly, at each discrete time step t, the state space (s_t) and action space (a_t) of the upper agent are defined as follows:

$$\begin{cases} s_{t,\text{upper}}=\{V_x,\ \Delta V_x,\ \beta,\ \Delta\beta,\ \gamma,\ \Delta\gamma,\ \delta_{\text{f}}\}, \\ a_{t,\text{upper}}=\{T_{\text{d}},\ M_{\text{d}}\}, \end{cases} \tag{5}$$

where $\Delta V_x=V_{xt}-V_x$ denotes the deviation between the target and actual longitudinal velocities; V_{xt} is the target longitudinal velocity; $\Delta\beta=\beta_{\mathrm{des}}-\beta$ represents the deviation between the desired and actual sideslip angles; $\Delta\gamma=\gamma_{\mathrm{des}}-\gamma$ denotes the deviation between the desired and actual yaw rates; δ_{f} is the front wheel steering angle; T_{d} is the desired total driving torque; M_{d} is the desired total yaw moment.

The selection of the reward function (r_t) plays a crucial role in agent training. The reward function for the upper agent is formulated as follows, and the detailed coefficient values (a_1–a_6) and the threshold settings of piecewise function H_V are provided in Section S2.2 of the ESM:

$$r_{t,\mathrm{upper}}=-\Big[(a_1\cdot\Delta V_x+a_2\cdot\Delta\beta+a_3\cdot\Delta\gamma)+(a_4\cdot\dot{T}_{\mathrm{d}}+a_5\cdot\dot{M}_{\mathrm{d}})+a_6\cdot H_V\Big]. \tag{6}$$

2.2.2 Lower agent for safe-energy cooperative optimization

The lower agent distributes the driving torque and yaw moment from the upper agent. Conventional methods rely on fixed weighting coefficients, which fail to adapt to complex tire–road variations. To address this, our lower agent uses DRL to generate dynamic weighting coefficients, enabling adaptive coordination between safety and energy saving.

The tire workload ratio serves as a critical indicator of vehicle stability, reflecting the utilization level of the friction circle. The tire workload ratio is defined as follows (Suzuki et al., 2014):

$$\sigma_i=\frac{\sqrt{F_{x_i}^2+F_{y_i}^2}}{\mu F_{z_i}}, \tag{7}$$

where μ is the adhesion coefficient.

The tire friction circle characterizes the utilization boundary of tire forces (Fig. 4). A smaller tire workload ratio σ_i indicates that the resultant of longitudinal and lateral tire forces lies farther from the friction circle boundary, thereby providing a greater stability margin. Consequently, the vehicle exhibits enhanced disturbance rejection capability during operation, leading to improved stability.

For the safety objective, we minimized the total tire workload ratio across all wheels to increase the adhesion margin and enhance stability. For energy saving, we considered two aspects: motor power losses, which depend on motor torque and speed, and tire slip energy losses, which arise from longitudinal and lateral slip. These three objectives were formulated as:

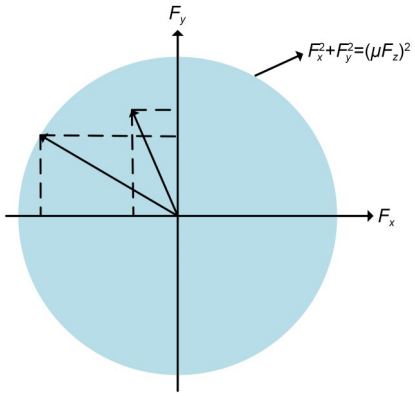


Fig. 4 Diagram of Kamm's circle for tire force/adhesion

$$\begin{cases} J_\sigma=\sum_{i=1}^{4}\sigma_i^2=\sum_{i=1}^{4}\dfrac{F_{x_i}^2+F_{y_i}^2}{(\mu F_{z_i})^2}, \\ J_{\mathrm{m}}=\sum_{i=1}^{4}E_{\mathrm{mloss},i}(T_{\mathrm{m},i},n_i), \\ J_{\mathrm{t}}=\sum_{i=1}^{4}(-F_{y_i}v_{x_i}\sin\beta_i+F_{x_i}\lambda_{x_i}v_{x_i}\cos\beta_i), \end{cases} \tag{8}$$

where J_σ, J_{m}, and J_{t} represent the tire workload ratio, motor loss, and tire slip loss, respectively; $E_{\mathrm{mloss},i}$ denotes the energy loss of the ith motor.

To comprehensively balance the dynamic coordination between active safety and energy-saving objectives, we normalized the total tire workload ratio, total tire slip energy consumption, and total motor losses, and combined them through a weighted summation, which denotes the sum of optimal values. The compact form of the proposed control model is expressed as follows:

$$J=\tau J_{\mathrm{safe}}+(1-\tau)J_{\mathrm{energy}}=\frac{\tau J_\sigma}{4}+\frac{(1-\tau)(J_{\mathrm{m}}+J_{\mathrm{t}})}{Q},$$

$$\text{s.t.}\begin{cases} \dfrac{l_{\mathrm{b}}}{2R_{\mathrm{eff}}}(T_{\mathrm{fr}}-T_{\mathrm{fl}}+T_{\mathrm{rr}}-T_{\mathrm{rl}})=M_{\mathrm{d}}, \\ T_{\mathrm{fl}}+T_{\mathrm{fr}}+T_{\mathrm{rl}}+T_{\mathrm{rr}}=T_{\mathrm{d}}, \\ -\sqrt{(\mu F_{z_i})^2-(F_{y_i})^2}\leqslant\dfrac{T_i}{R_{\mathrm{eff},i}}\leqslant\sqrt{(\mu F_{z_i})^2-(F_{y_i})^2}, \\ T_{i,\min}\leqslant T_i\leqslant T_{i,\max}, \end{cases} \tag{9}$$

where J represents the total coordinated control objective; J_{safe} and J_{energy} denote the safety objective and energy-saving objective, respectively; τ is the weighting coefficient; R_{eff} denotes the effective rolling radius of the wheel; T_{fl}, T_{fr}, T_{rl}, and T_{rr} denote the driving torques of the front-left, front-right, rear-left, and rear-right wheels, respectively; $T_{i,\min}$ and $T_{i,\max}$ denote its lower and upper torque limits, respectively. For the in-wheel motor configuration considered in this study, T_i is equivalent to the motor torque $T_{\text{m},i}$ defined in Eq. (3). Q denotes the sum of optimal motor power losses and tire slip power losses under the current vehicle state; s.t. points to the constraint set, sequentially comprising the yaw-moment control constraint, total driving torque constraint, road adhesion constraint, and motor output constraint.

The weighting coefficient τ in Eq. (9) is not a fixed constant preset by the designer, but rather an action output generated online by the lower-level agent's policy network at each time step based on the current vehicle state. The value of $\tau\in[0, 1]$ dynamically adjusts the dominance between the safety objective and the energy-saving objective in the comprehensive optimization. When the vehicle experiences a large sideslip angle or yaw rate deviation, the agent tends to output a larger τ to prioritize safety, whereas when the vehicle operates in a stable region, the agent tends to output a smaller τ to optimize energy efficiency. This adaptive trade-off mechanism constitutes the core advantage of the HDRL approach.

The weighting coefficient τ serves as the output action of the lower agent. To provide the agent with sufficient critical information for decision-making, the state space and action space of the lower agent are defined as follows:

$$\begin{cases} s_{t,\text{lower}}=\Big\{T_{\text{d}},\ M_{\text{d}},\ V_x,\ \Delta V_x,\ \int\Delta V_x,\ \beta,\ \Delta\beta,\\ \qquad\qquad \int\Delta\beta,\ \gamma,\ \Delta\gamma,\ \int\Delta\gamma,\ \delta_{\text{f}}\Big\},\\ a_{t,\text{lower}}=\{\tau\}. \end{cases} \tag{10}$$

The reward function for the lower agent is formulated as follows:

$$r_{t,\text{lower}}=-(b_1\cdot P_{\text{m}_{\text{loss}},\text{total}}+b_2\cdot P_{\text{t}_{\text{loss}},\text{total}}+b_3\cdot H_\gamma+b_4\cdot H_\beta), \tag{11}$$

where b_1–b_4 are weighting coefficients; $P_{\text{m}_{\text{loss}},\text{total}}$ denotes the total power loss of the four in-wheel motors; $P_{\text{t}_{\text{loss}},\text{total}}$ denotes the total tire slip power loss of the four wheels; H_β and H_γ are piecewise penalty functions for sideslip angle and yaw rate errors, respectively. The detailed definitions of these piecewise functions, along with the threshold determination process and the final coefficient values, are provided in Section S2.3 of the ESM.

After the lower agent outputs the weighting coefficient τ, QP is used to solve the optimization problem. The driving torques of the four wheels are defined as the decision variable vector $\boldsymbol{u}=[T_{\text{fl}},\ T_{\text{fr}},\ T_{\text{rl}},\ T_{\text{rr}}]^{\text{T}}$. Based on the integrated objective function in Eq. (9), and incorporating the quadratic characteristics of the tire workload ratio together with the comprehensive energy consumption model, as defined in Eq. (8), the objective function in standard QP form is formulated as follows:

$$\min_{\boldsymbol{u}} J(\boldsymbol{u})=\frac{1}{2}\boldsymbol{u}^{\text{T}}\boldsymbol{H}\boldsymbol{u}+\boldsymbol{f}^{\text{T}}\boldsymbol{u}, \tag{12}$$

where $\boldsymbol{H}$ denotes the positive definite Hessian matrix; $\boldsymbol{f}$ represents the linear term vector, incorporating the first-order coefficients from the comprehensive energy consumption model. The objective function is subject to equality constraints on the total driving force and yaw moment, and inequality constraints on the motor torque limits and tire–road adhesion. The detailed constraint expressions and solver settings are provided in Section S2.4 of the ESM.

2.2.3 State and action preprocessing

All state variables are linearly normalized based on predefined physical bounds to ensure consistent gradient magnitudes during training. The action spaces are constrained as follows:

$$\begin{cases} T_{\text{d}}\in[-4T_{\text{m,max}},\ 4T_{\text{m,max}}],\\ M_{\text{d}}\in[-M_{\max},\ M_{\max}],\\ \tau\in[0,\ 1], \end{cases} \tag{13}$$

where the first two bounds are derived from the motor peak torque and track width, and the last is enforced via a sigmoid activation function. These preprocessing strategies improve training stability, accelerate convergence, and prevent infeasible action outputs.

2.2.4 SAC and network architecture design

As previously discussed, in the domain of DRL, the SAC algorithm has shown a strong capability in exploring complex policy spaces for multi-objective optimization tasks and has been widely adopted. Within SAC, the actor uses a policy-based approach to learn the policy π, outputting agent actions to maximize cumulative rewards obtained from the environment, while the critic uses a value-based approach to learn value functions for evaluating the current state or actions output by the actor (Konda and Tsitsiklis, 1999). The SAC algorithm is built upon the MDP framework introduced in Section S2.1, with its detailed mathematical formulations provided in Section S2.5 of the ESM.

In the SAC algorithm, both the actor and critic networks are implemented as fully connected neural networks (FCNNs) with rectified linear unit (ReLU) activation functions. The network architectures are illustrated in Fig. 5.

3 Results and discussion

The proposed control method was implemented and validated on the CarSim–Simulink co-simulation platform. Parameters of the vehicle and the HDRL controller are summarized in Tables S2 and S3 of the ESM, respectively. The weighting coefficients in the reward functions are listed in Table S4 of the ESM. Training was conducted under double lane change (DLC) maneuvers performed on a wet low-adhesion road surface (μ=0.3) with vehicle longitudinal speeds ranging from 60 to 80 km/h. Training was conducted on a workstation featuring 16 GB of random-access memory and an AMD Ryzen 7 2700X octa-core central processing unit (CPU), and implemented using MATLAB's reinforcement learning toolbox. To validate the benefits of the proposed hierarchical architecture over a non-hierarchical single-agent DRL control method in terms of accelerated training convergence and reduced optimization difficulty, the training processes of both approaches were compared. The single-agent DRL method directly outputs four torque control variables through a single agent, using actor and critic network architectures identical to those in the HDRL framework and a reward function based on the summed rewards of the two HDRL agents. The training reward curves are presented in Fig. 6. The single-agent DRL method failed to achieve convergence within the first 900 episodes, whereas the upper agent of the HDRL framework gradually converged after about 700 iterations, and the lower agent converged after about 500 iterations. During weight tuning, any combinations that led to training divergence or vehicle instability were discarded. The final set of weights adopted after this screening consistently yielded convergent performance across multiple repeated training runs. This demonstrates that the HDRL control system progressively stabilized with optimized control performance, exhibiting substantially improved training stability and convergence speed compared to the single-agent DRL approach.

3.1 Case 1: 2D road validation

To comprehensively validate the effectiveness and generalization capability of the proposed control

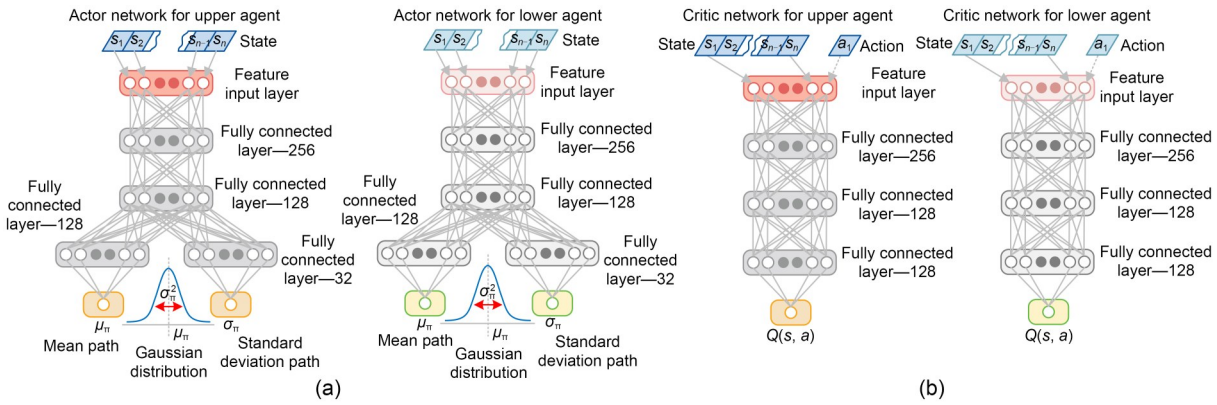


Fig. 5 SAC network architecture: (a) actor network design; (b) critic network design. μ_π denotes the mean and σ_π denotes the standard deviation of Gaussian distribution; $Q(s, a)$ denotes the state-action value function, which evaluates the expected cumulative return of taking action a in state s

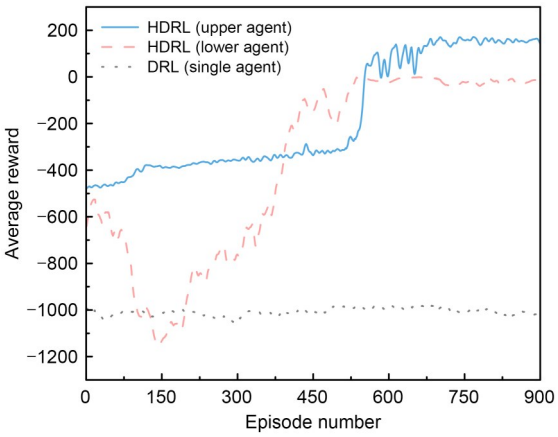


Fig. 6 Training process

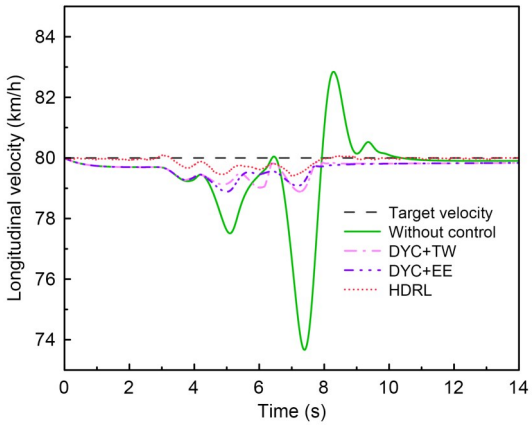


Fig. 7 Velocity in DLC maneuver

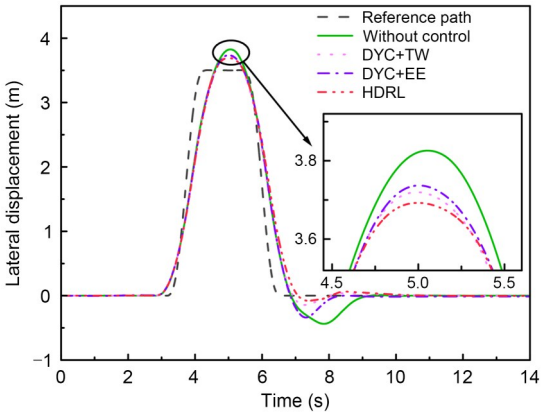


Fig. 8 Lateral displacement in the DLC maneuver

method, the response performance of the trained agents was evaluated under two maneuvers, DLC and step steering input. Three baseline methods were selected for comparison: (1) Without control, i.e., equal distribution of driving torque among wheels without yaw stability control; (2) DYC+TW (direct yaw-moment control considering tire workload), i.e., torque allocation minimizing the tire workload ratio with yaw-moment control, prioritizing vehicle stability through full utilization of tire adhesion limits; (3) DYC+EE (direct yaw-moment control considering energy efficiency), i.e., torque allocation minimizing total power losses (tire slip and motor losses) with yaw-moment control, optimizing energy efficiency without explicit consideration of safety metrics such as tire workload ratio. Following conventional practice from previous studies, in both DYC+TW and DYC+EE, a PID controller was used to regulate longitudinal speed and a sliding-mode controller was used to generate the required yaw moment. The proposed method is denoted as HDRL. As the single-agent DRL control method failed to achieve training convergence (Fig. 6), its results were excluded from the comparative experiments.

The DLC maneuver serves as an effective test for simulating overtaking scenarios and emergency avoidance maneuvers. The vehicle was driven at a target longitudinal speed of 80 km/h on a wet road surface with a low adhesion coefficient of μ=0.4. The results are presented in Figs. 7–11.

Yaw rate responses under different control methods are illustrated in Fig. 9. The proposed HDRL method exhibited the best yaw rate tracking performance, with a maximum absolute yaw rate of about 0.38 rad/s and a maximum deviation from the reference value of 0.19 rad/s. In comparison, the equal driving torque distribution method without yaw stability control (Without control) yielded a maximum absolute yaw rate of 0.78 rad/s and a maximum deviation of 0.60 rad/s; the DYC+EE method yielded 0.51 and 0.32 rad/s, and the DYC+TW method yielded 0.41 and 0.23 rad/s, respectively. These results indicate that HDRL achieves superior yaw dynamics regulation with significantly reduced overshoot and tracking error compared to all baseline methods. HDRL constrained the sideslip angle below 0.10 rad, outperforming DYC+EE (0.14 rad), DYC+TW (0.15 rad), and Without control (0.48 rad) (Fig. 10). Regarding energy loss (Fig. 11), HDRL achieved 48.24 kJ, second only to DYC+EE (48.05 kJ), while Without control and DYC+TW incurred losses of 91.22 and 58.04 kJ, respectively. These results confirm that HDRL delivers effective energy saving without compromising safety.

The radar chart in Fig. 12 further shows that HDRL achieved superior holistic performance in

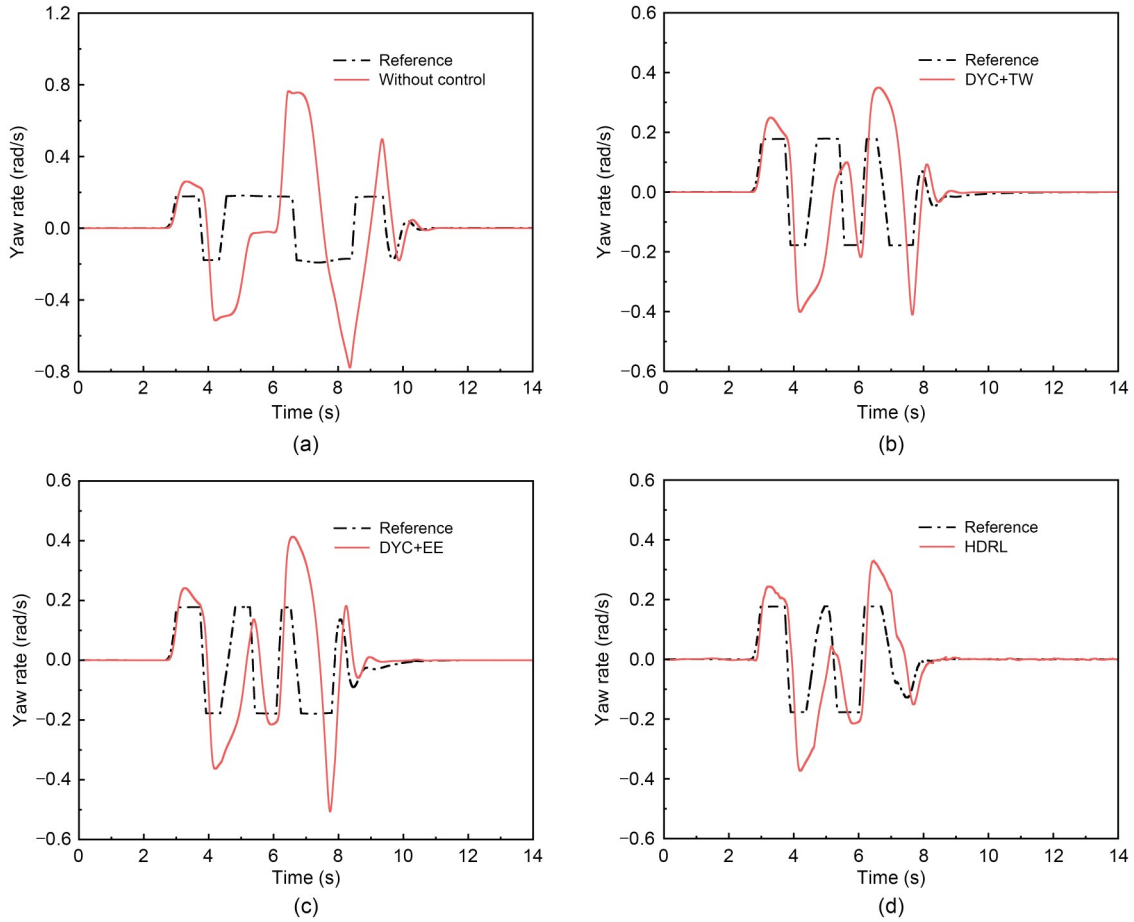


Fig. 9 Yaw rate in the DLC maneuver: (a) Without control; (b) DYC+TW; (c) DYC+EE; (d) HDRL

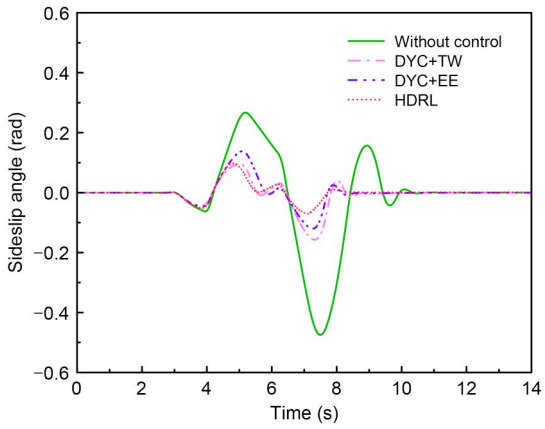


Fig. 10 Sideslip angle in the DLC maneuver

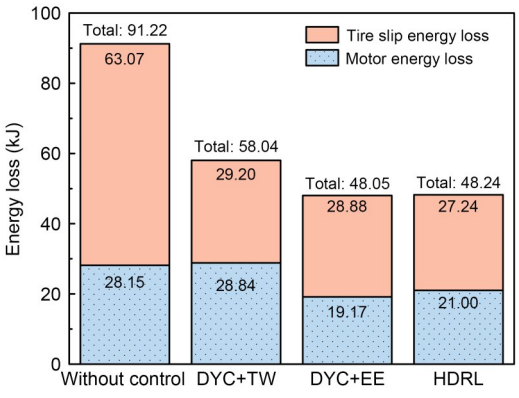


Fig. 11 Energy loss in the DLC maneuver

multi-objective coordinated optimization compared to all baselines.

To verify the generalization capability, a step steering maneuver was conducted at 80 km/h on a dry road (μ=0.8), with a 39° step command applied at 3 s. HDRL achieved the highest longitudinal speed tracking accuracy (with an error less than 0.35 km/h), a maximum yaw rate error of about 0.03 rad/s, and a sideslip angle error of about 0.03 rad (Figs. 13–16). Its total energy loss ranked second only to DYC+EE. These results confirm that HDRL delivers favorable coordinated control of safety and energy saving with robust generalization across diverse driving conditions.

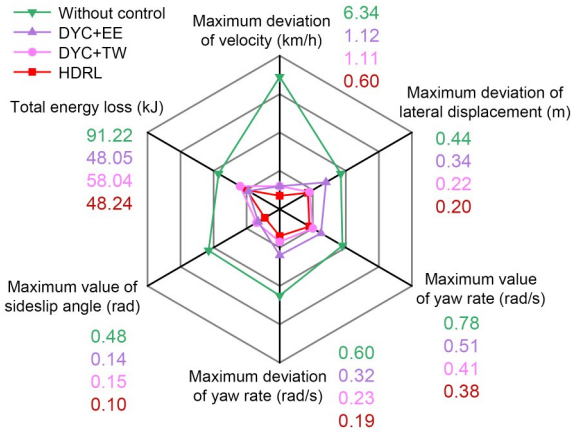


Fig. 12 Radar chart of the comprehensive performance

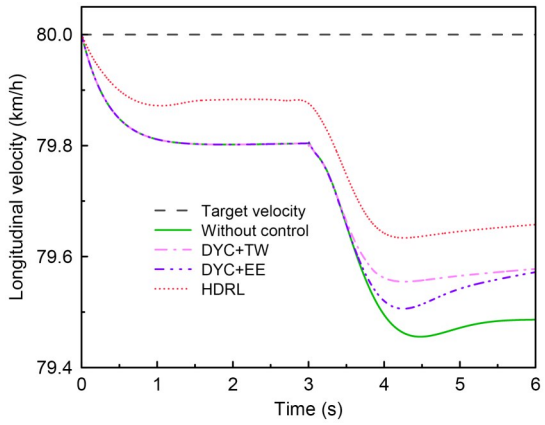


Fig. 13 Velocity in the step steering maneuver

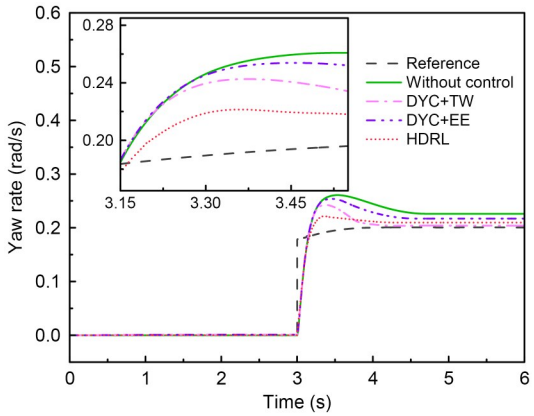


Fig. 14 Yaw rate in the step steering maneuver

3.2 Case 2: 3D road validation

To further evaluate the proposed method under realistic road conditions, we conducted extended validation on a 3D featured road segment. A segment of about 1.1 km exhibiting continuous elevation variations

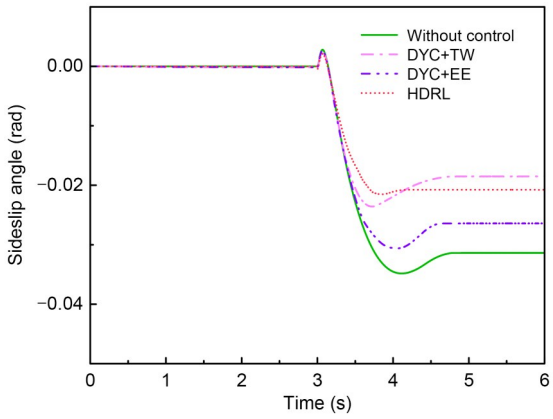


Fig. 15 Sideslip angle in the step steering maneuver

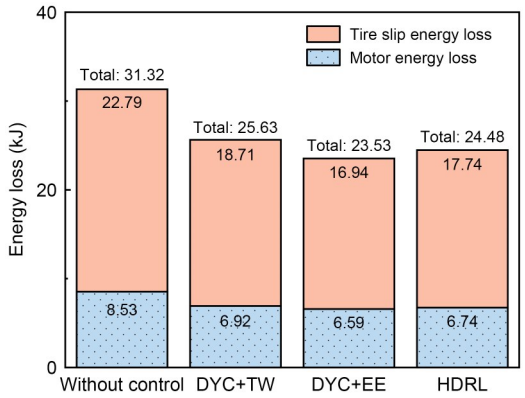


Fig. 16 Energy loss in the step steering maneuver

was selected as the validation scenario (Fig. S2 of the ESM). It had a maximum upgrade of 3.89%, a maximum downgrade of −3.28%, and an average absolute gradient of 1.21%. The detailed road geometry, including elevation profiles, gradient distribution, and curvature variations, is shown in Fig. S3 of the ESM.

The vehicle was tested at 50 km/h on dry asphalt (μ=1.0) over a 360 s (about 5 km) trajectory. Two baselines were compared: (1) Without control (equal torque distribution and no yaw control); (2) LQR (linear quadratic regulator; yaw-moment control with load-proportional torque allocation). The results are presented in Figs. 17–20.

As shown in Figs. 17–19, HDRL achieved the smallest speed deviation (max error 0.16 km/h, a reduction of 0.66 km/h vs. Without control), the smallest sideslip angle (≈0.04 rad, vs. 0.05 rad for LQR and >0.1 rad for Without control), and the most accurate yaw rate tracking. These results confirm that HDRL effectively enhances both longitudinal and lateral stability.

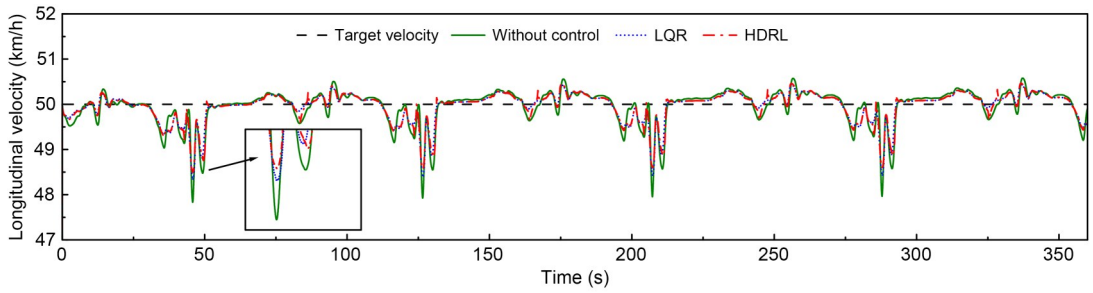


Fig. 17 Velocity in the 3D road validation

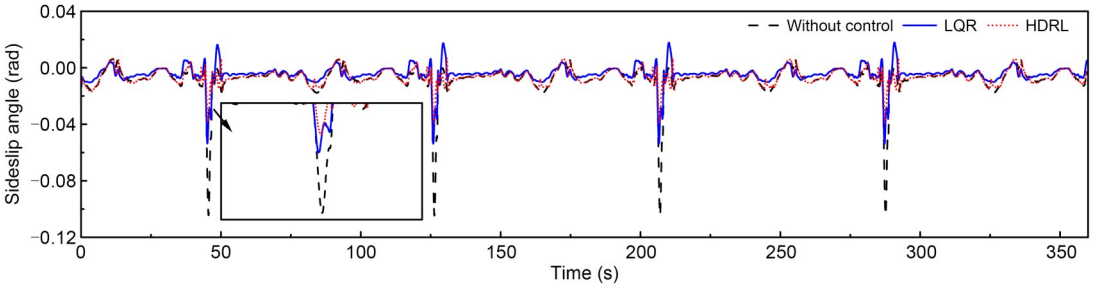


Fig. 18 Sideslip angle in the 3D road validation

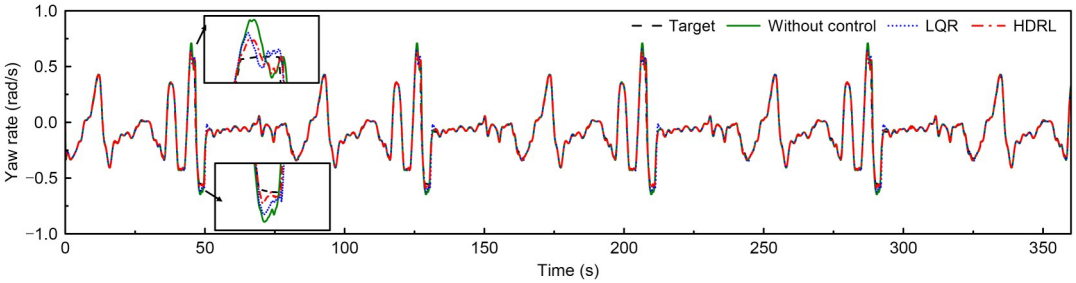


Fig. 19 Yaw rate in the 3D road validation

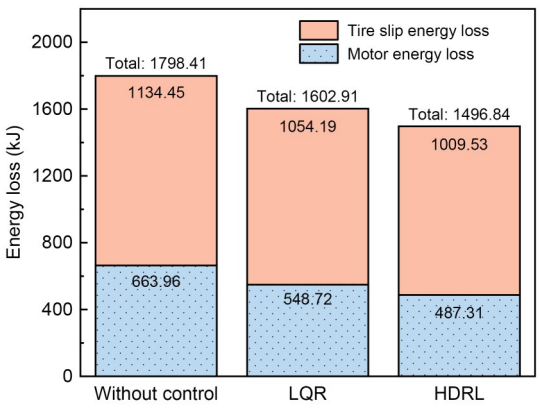


Fig. 20 Energy loss in the 3D road validation

Wheel torque distribution profiles under different control methods are shown in Fig. S4 of the ESM. In summary, the HDRL controller concentrated torque within the high-efficiency region (200–450 N·m), whereas baseline strategies operated mainly in the low-efficiency range (0–100 N·m). This distribution pattern reflects the HDRL agent's learned capability for multi-objective coordinated optimization, as it prioritizes stability during critical maneuvers and energy efficiency when stability margins are sufficient.

Energy saving represents another critical consideration in controller design. In this section, the proposed HDRL strategy is evaluated based on three metrics: total energy loss, tire slip energy loss, and motor energy loss. Over the 5-km test trajectory, the HDRL strategy achieved a total energy loss of only 1496.84 kJ (Fig. 20). In comparison, the Without control and LQR methods incurred substantially higher losses of 1798.41 and 1602.91 kJ, respectively. Consequently, the HDRL approach yielded energy savings of 6.62% to 16.77%

relative to the baseline strategies, demonstrating its superior efficiency in coordinated energy management.

3.3 Case 3: real-time hardware-in-the-loop validation

To assess the real-time feasibility and deployment potential of the proposed HDRL method, real-time hardware-in-the-loop (HIL) validation was conducted. The HIL platform comprised a host computer and a real-time target machine (Fig. 21). The host computer served as the central development and configuration unit, where control algorithms were designed and simulation models constructed, and the real-time target was configured using MATLAB/Simulink. Communication between the host and target was established via Transmission Control Protocol/Internet Protocol (TCP/IP). The real-time target machine was a Speedgoat Performance unit equipped with a 4.2 GHz quad-core CPU and an optional Simulink-programmable field programmable gate array (FPGA), delivering robust real-time execution capability. The specifications of the real-time target machine are summarized in Table 1. To evaluate robustness under parametric uncertainty, a vehicle mass perturbation test was designed wherein the total vehicle mass was varied by ±10% relative to its nominal value, to simulate payload variations. The vehicle executed a DLC maneuver at a target longitudinal speed of 72 km/h on a wet low-adhesion road surface (μ=0.3). HIL results were compared against simulation (SIM) results obtained under nominal mass conditions. The corresponding outcomes are presented in Figs. 22–26.

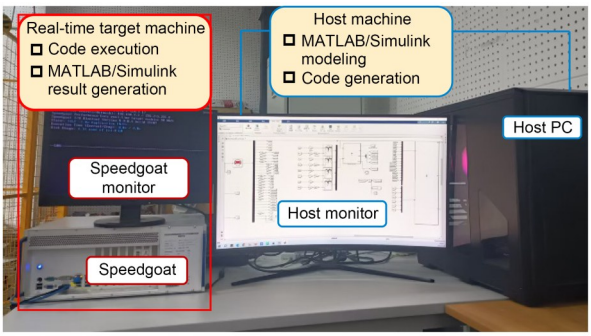


Fig. 21 HIL experimental configuration for real-time validation

Table 1 Detailed specifications of the real-time target machine

Specification	Description
Model	DH-PR-TTM-4.2
Processor	Intel Core i7 4.2 GHz, 4 cores
Memory	4 GB
Main drive	1 TB

As shown in Figs. 22 and 23, the HDRL controller maintained longitudinal velocity tracking within acceptable deviation bounds under mass perturbations in the

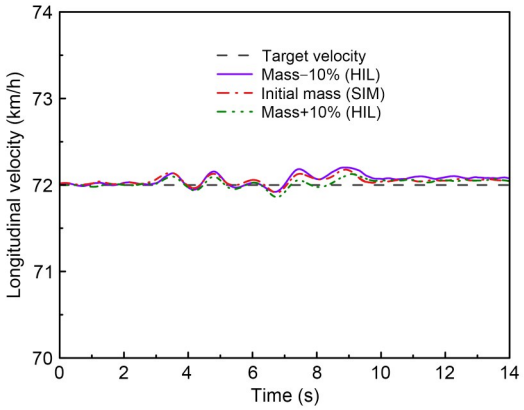


Fig. 22 Velocity in robustness validation

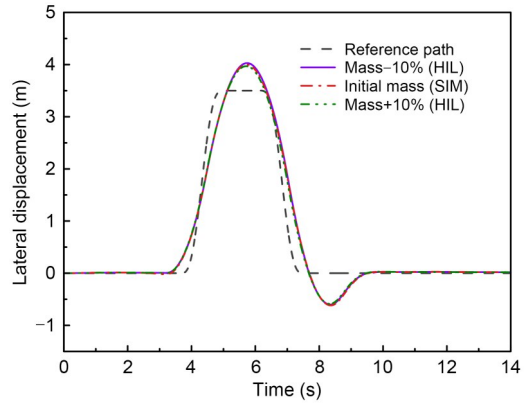


Fig. 23 Lateral displacement in robustness validation

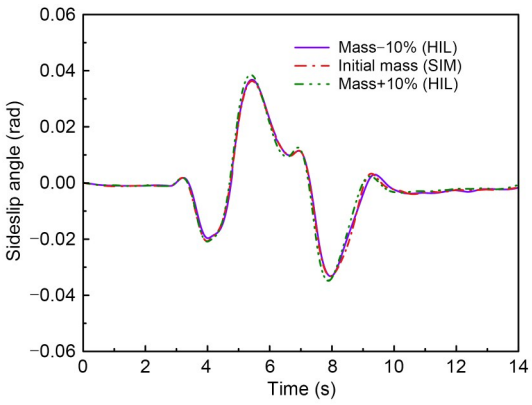


Fig. 24 Sideslip angle in robustness validation

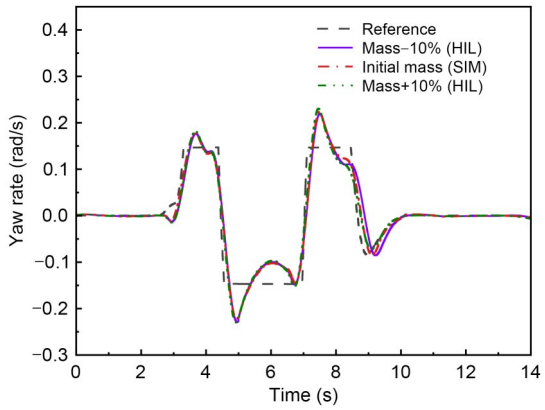


Fig. 25 Yaw rate in robustness validation

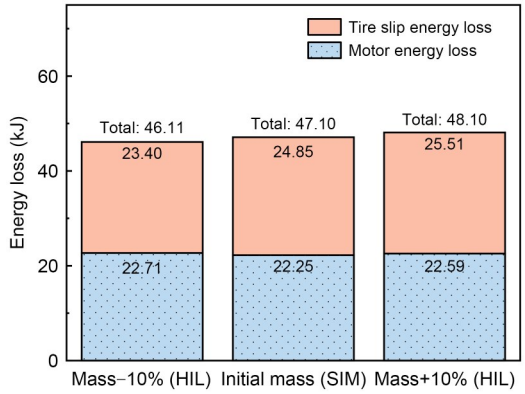


Fig. 26 Energy loss in robustness validation

HIL environment. Furthermore, Figs. 24–26 reveal that under HDRL control, both the peak values and temporal trajectories of the yaw rate and sideslip angle remained largely unaffected by vehicle mass variation. Concurrently, the energy saving performance exhibited minimal sensitivity to mass perturbations, with deviations confined within 2%. These results underscore the robustness of the HDRL framework in maintaining consistent dynamic regulation and energy efficiency under parametric uncertainty.

Consequently, the HIL and SIM results showed highly consistent performance with no observable systematic discrepancies. These experiments validate the robustness of the proposed HDRL method. Exploiting the nonlinear approximation capability of DRL policy networks, the controller implicitly learns compensation mechanisms for parametric variations, thereby maintaining stable control performance and real-time responsiveness under perturbations. This robustness confirms the suitability of the HDRL framework for in-vehicle deployment.

4 Conclusions

This study quantified tire slip losses and integrated them with motor losses into a unified multi-objective framework as part of a proposed HDRL architecture for IWMs that coordinates active safety and energy saving. The upper agent manages longitudinal speed and yaw stability, while the lower agent dynamically balances safety and efficiency via adaptive weighting. The main findings were as follows:

(1) Hierarchical decomposition substantially improves training stability and convergence over single-agent DRL, validating the framework's ability to handle the safety–efficiency trade-off in IWMs.

(2) Under DLC on low-adhesion surfaces, HDRL achieved a lateral displacement error of 0.20 m, a yaw rate peak of 0.38 rad/s (deviation 0.19 rad/s), and a sideslip angle below 0.10 rad—matching the safety-oriented DYC+TW. The total energy loss was 48.24 kJ, second only to DYC+EE (48.05 kJ), while providing substantially better safety. Consistent performance in step steering confirmed robustness across transient and steady-state conditions.

(3) On a 5-km 3D road with elevation changes, HDRL reduced the maximum speed deviation by up to 0.66 km/h, constrained sideslip angle below 0.04 rad, and achieved the lowest yaw rate error, while cutting total energy losses by 6.62%–16.77%.

(4) HIL tests under ±10% mass perturbations showed that HDRL maintained stable dynamic responses across all metrics, with energy-saving deviations within 2%, confirming its robustness to parametric uncertainty.

In summary, the proposed HDRL method delivers effective coordinated control of active safety and energy saving for IWMs under diverse operating conditions and uncertainties.

In future work, we will explore integrating vehicle dynamics state observations with online parameter identification to improve controller generalization and robustness, and investigate an edge–cloud collaborative evolution framework to validate real-world deployment performance on production vehicles.

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Author contributions

Yalei LIU designed the study; Yalei LIU and Xinyu QI conceived and conducted the simulations and experiments; Yalei LIU and Honglin ZHU analyzed the experimental results; Yalei LIU drafted the paper; Mingliang YANG and Weiping DING provided insights on the draft; Yalei LIU and Xinyu QI revised the draft; all authors reviewed the final manuscript.

Conflict of interest

Yalei LIU, Xinyu QI, Weiping DING, Honglin ZHU, and Mingliang YANG declare that they have no conflicts of interest.

Declaration on the use of generative AI tools

During the preparation of this work, the authors used DeepSeek and Grammarly to improve language and readability. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Electronic supplementary materials

Sections S1–S3, Tables S1–S4, Figs. S1–S4